

TURRET OS: Multi-Layered Graph and Rule-AST Architecture Rescuing Insider Threat Detection Against Out-of-Distribution Adversaries

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ABSTRACT

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1. Introduction
Insider incidents—abuse of authorised access by legitimate users—continue to be a leading operational, financial, and regulatory exposure for enterprise security programmes. The dominant defensive response today is machine-learning detection over user-behaviour features, file-system telemetry, and provenance graphs; however, when reduced to a single feature layer, these detectors degrade sharply under out-of-distribution adversarial adjustment, and their alert outputs rarely satisfy courtroom admissibility standards. This paper addresses two open problems in tandem: detector resilience under held-out out-of-distribution adversaries, and a continuous inference-to-evidence chain that delivers forensically admissible artefacts.

Prior research clusters in four strands. The first applies graph-neural architectures to multi-modal insider telemetry—SENTINEL [Xiao et al., 2024], DTGI [Gao et al., 2023], MEWRGNN [Xiao et al., 2022], and TGCN-DA [Li et al., 2023]—alongside feature-engineered baselines including Le and Zincir-Heywood [Le et al., 2020] and E-Watcher [Wei et al., 2024]. The second strand builds provenance-graph cybersecurity for tactical attack attribution—CSKG4APT [Ren et al., 2022], KRYSTAL [Kurniawan et al., 2022], KGRAG [Aki et al., 2025], and the provenance-graph survey of Li et al. [Li et al., 2021]—but typically terminates outputs at alert generation. The third strand extracts file-metadata signals: image-EXIF vectorisation in Exif2Vec [Umair et al., 2024] and Yasenenko [Yasenenko, 2025], and OOXML-revision and tamper forensics in Didriksen [Didriksen, 2014] and Fu et al. [Fu et al., 2015]. The fourth strand formalises digital forensic readiness under ISO/IEC 27043 for SaaS insider threats [Shoderu et al., 2025, 2026] and grounds provenance representation in W3C PROV [Yusof et al., 2025, Sikos and Philp, 2020] and operational trace frameworks [Garfinkel et al., 2013].

Several gaps remain. (i) Published detectors are predominantly single-layer—feature-engineered pipelines, monolithic graph neural networks, or stand-alone rule engines—and degrade sharply when faced with out-of-distribution adversaries whose observable features converge with benign

baseline traffic [Wei et al., 2024, Xiao et al., 2024, Gao et al., 2023]. (ii) File-metadata work remains narrowly restricted to image EXIF; cross-format provenance covering OOXML revisions, DWG authorship, EML headers, and git commit chains is absent [Umair et al., 2024, Yasenenko, 2025, Didriksen, 2014, Fu et al., 2015]. (iii) Provenance-graph research emits alerts but rarely couples them to W3C-PROV-aligned evidence chains suitable for forensic-readiness audits [Ren et al., 2022, Kurniawan et al., 2022, Aki et al., 2025, Li et al., 2021]. (iv) Readiness frameworks under ISO/IEC 27043 target SaaS-cloud deployments and have not been extended to on-premise or hybrid enterprise environments [Shoderu et al., 2025, 2026]. (v) No prior system braids multi-layer inference, cross-format metadata extraction, and court-grade evidence delivery in a single architecture.

Against these gaps we propose TURRET OS,¹ a four-layer architecture that fuses a cross-format metadata-harvest layer (L1), a Neo4j knowledge-graph layer (L2), a rule-AST inference layer (L3), and a multi-layer GraphSAGE inference layer (L3.5) into one calibrated decision surface; a fourth layer emits W3C-PROV-grounded, Merkle-chain-integrity-protected, Ed25519-signed evidence packs aligned to all eight ISO/IEC 27043 readiness attributes—Identification, Collection, Acquisition, Preservation, Analysis, Presentation, Chain of Custody, and Integrity Verification. Under an out-of-distribution credential-phishing adversary the multi-layer stack retains AUC 0.9572 with only a 4.12% drop, while single-layer feature detectors collapse to AUC 0.7363 with a 26.25% drop—a +22.09% architectural rescue. On the D5 Synthetic-Tenant Pilot (35 users, 90 days) with a held-out OOD profile, full-stack detection attains ROC-AUC 0.9928, F1 @ 0.5% FPR of 0.8936, and MCC 0.9539; the top SHAP feature is `copy_to_removable`, an operationally interpretable signal verified by a runtime feature-whitelist invariant that guarantees zero label leakage at deployment. End-to-end ISO/IEC 27043 readiness coverage is verified at 100% across all eight required attributes.

The remainder of the paper is organised as follows. Section 2 reviews prior literature across graph-neural insider detection, provenance-graph cybersecurity, file-metadata

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¹Project repository: <https://github.com/ali-Hamza817/Turret-OS>

forensics, and ISO/IEC 27043 readiness. Section 3 specifies the adversary model, attack vectors, and evaluation criteria. Section 4 presents the TURRET OS architecture layer by layer. Section 5 reports the implementation stack on an 80-core, 503.58 GB-RAM, NVIDIA L40S, Python 3.12.9 platform. Section 6 details the experimental evaluation, including baseline comparison, layer ablation study, Pareto PR frontier, single-layer versus multi-layer adversarial robustness, ISO/IEC 27043 coverage, and compute budget. Section 7 discusses architectural and operational implications. Section 8 outlines limitations and future work. Section 9 concludes.

2. Literature Review

Earliest work on insider-threat detection framed the attacker as legitimate trusted credential misuse and pursued identification through hand-crafted rule heuristics and lightweight media forensics. [Garfinkel et al. \[2013\]](#) proposed media-based detection by content-aware fingerprinting and metadata-aware surface analysis of seized artefacts, demonstrating that visible media artefacts alone can discriminate threatening from benign insider behaviour. [Didriksen \[2014\]](#) examined the forensic artefacts of OOXML documents produced by Microsoft Office, showing that revision identity, custom-XML fingerprints, and document-version metadata can be assimilated into identification pipelines for document-level insider analysis. [Fu et al. \[2015\]](#) extended this line by analysing covert channels in Office 2007–2013 documents, demonstrating that XML structure and revision metadata can leak secrets to a co-leaked artefact. Limitations of this early media-only line were threefold: (i) coverage was restricted to a small set of document formats; (ii) operationally deployable detection was not demonstrated; and (iii) no automatic chain of forensic evidence was raised from the artefacts.

The next research wave widened the problem definition and laid the foundations of multi-modal detection and provenance representation. [Mavroeidis et al. \[2018\]](#) introduced a hybrid data-driven physical-security framework merging anomaly detection with user-behaviour profiling, demonstrating improvement of baseline detection on enterprise telemetry. [Le and Zincir-Heywood \[2019\]](#) and [Le et al. \[2020\]](#) systematically compared ensemble-ML and deep-learning approaches across multiple data-granularity levels of CERT insider datasets, with their longitudinal studies reporting machine-learning detection at AUROC 0.92 and F1 0.88 with time-to-detect of 14 minutes. [Homoliak et al. \[2018\]](#) catalogued the taxonomy of insider threats, attack vectors, and defensive controls in their seminal *Insight Into Insiders and IT*, establishing a reference map for the subsequent ML-detection lineage. [Sikos and Philp \[2020\]](#) surveyed provenance-aware knowledge representation models and contextualised knowledge graphs, advancing the formal foundations for representing and reasoning over cyber provenance traces. [Li et al. \[2021\]](#) consolidated system-level provenance-graph survey work for threat detection, establishing that the provenance-graph paradigm is the natural

substrate for post-hoc forensic reasoning over multi-modal telemetry. The limitation of this generation is structural: detection and provenance were treated as separate worlds, no system delivered both with court-grade evidence, and adversarial robustness was not yet measured through held-out attacker profiles.

The third research wave tied provenance-graph analytics to detection of advanced threats and assessed adversarial robustness experimentally. [Ren et al. \[2022\]](#) constructed CSKG4APT—a cybersecurity knowledge graph for advanced persistent threat organisation attribution—by fusing indicators of compromise, threat reports and malware artefacts into a unified graph. [Kurniawan et al. \[2022\]](#) introduced KRYSTAL—a knowledge-graph framework for tactical attack discovery in audit data—demonstrating graph-rule fusion for tactical-stage detection. [Xiao et al. \[2022\]](#) introduced MEWRGNN, a graph-neural-network architecture for anomaly-based insider-threat detection reporting AUROC 0.94 and F1 0.87. [Gao et al. \[2023\]](#) attacked the class-imbalance problem with Deep Temporal Graph Infomax and reported AUROC 0.91 and F1 0.86. [Li et al. \[2023\]](#) proposed TGCN-DA, a temporal graph convolutional network with data augmentation, and reported AUROC 0.95 and F1 0.89. [Xiao et al. \[2024\]](#) advanced the lineage with SENTINEL, a multi-timescale user-behaviour interaction graph learner reporting AUROC 0.93 and F1 0.88. [Wei et al. \[2024\]](#) reported E-Watcher with 98.48% accuracy and 99.23% F1 on CERT r4.2 on a 64 GB MacBook M1 Max platform, securing the most robust reported figures of the lineage but omitting adversarial robustness, cross-format provenance, and forensic-readiness attributes. [Mitra et al. \[2021\]](#) proposed provenance-based counter-measures against fake cyber-threat intelligence, demonstrating that provenance can guard the quality of processed alerts. The limitation of this generation is that detection capability and provenance representation deepened separately; adversarial robustness was not yet measured through held-out attacker profiles, no cross-format file-metadata vector was exploited, and operational evidence chains remained disconnected from ML alerts.

The most recent wave began treating metadata, provenance, and forensic-readiness as first-class research targets. [Umair et al. \[2024\]](#) proposed Exif2Vec, a framework to ascertain untrustworthy crowdsourced images by vectorising EXIF metadata, demonstrating that image-level provenance signals can expose tampering behaviour. [Yasenko \[2025\]](#) evaluated image-recognition systems for automatic detection of malicious files based on image metadata, reporting AUROC 0.76 on a curated image set. [Aki et al. \[2025\]](#) extended KGRAG—provenance-based cyber-threat-intelligence analysis with knowledge graphs and retrieval-augmented generation—pushing provenance representation into LLMs. [Yusof et al. \[2025\]](#) introduced PROVCON, an observability-to-provenance framework, demonstrating that operational telemetry can be elevated to W3C PROV provenance. [Shoderu et al. \[2025\]](#) and [Shoderu et al. \[2026\]](#) formalised digital forensic readiness under ISO/IEC 27043 for

the SaaS-cloud environment, but their evaluation was scoped to SaaS-cloud deployments and did not couple readiness to ML detection or on-premise/hybrid enterprise settings [Shoderu et al., 2025, 2026]. The remaining open gap of this generation is the absence of a unifying architecture that fuses cross-format metadata extraction, multi-layer joint inference, ISO/IEC 27043 readiness, and held-out attacker profiles within a single deployable system.

Synthesising the limitations of the four research waves—narrow format coverage of media forensics, separation of detection and provenance, absence of held-out adversarial regimes, and missing on-premise forensic-readiness coupling—we propose TURRET OS,² a four-layer architecture that fuses cross-format metadata harvest (L1), a Neo4j knowledge graph (L2), a rule-AST inference engine (L3), and multi-layer GraphSAGE inference (L3.5) into one calibrated decision surface; a fourth layer emits W3C-PROV-grounded, Merkle-chained, Ed25519-signed evidence packs aligned to all eight ISO/IEC 27043 readiness attributes. End-to-end detection with the multi-layer stack attains ROC-AUC 0.9997, F1 @ 0.5% FPR of 0.8774, and MCC 0.9770 on the D3 TSEC corpus; under an out-of-distribution credential-phishing adversary the multi-layer stack retains AUC 0.9572 with only 4.12% drop, while single-layer L1 feature detectors collapse to AUC 0.7363 with 26.25% drop—a +22.09% architectural rescue. On the D5 Synthetic-Tenant Pilot (D5, 35 users over 90 days, proactively tagged by Security Operations analysts) with a held-out OOD profile, full-stack detection attains ROC-AUC 0.9928, F1 @ 0.8936, and MCC 0.9539; SHAP feature-importance audit with a runtime whitelist invariant guarantees zero label leakage at deployment, with `copy_to_removable` as the dominant operational signal. ISO/IEC 27043 readiness coverage is verified at 100% across all eight attributes; computational budget is profiled with wall-clock timestamps on an 80-core, 503.58 GB-RAM, NVIDIA L40S, Python 3.12.9 deployment platform. TURRET OS thus closes the four limitations jointly, advancing the state-of-the-art from incremental single-layer detectors to a single deployable braided architecture with court-grade forensic evidence delivery.

3. Threat Model and Problem Formulation

3.1. Adversary Profiles

TURRET OS evaluates adversary profiles across two corpora. On the established CERT-style insider-threat lineage corpus (D3 TSEC) the embedded adversary baseline consists of five synthetic insider scenarios generated over 500 users, 365 days, and 141,657 user-day activity records, with a 1.68% positive rate [Wei et al., 2024, Xiao et al., 2024, Gao et al., 2023]. On the 35-user, 90-day D5 Synthetic-Tenant Pilot (2,431 activity records) a held-out OOD_LAB_EXFILTRATOR profile is injected into the generator without appearing in the training split; this deliberately out-of-distribution scenario functions as a true hold-out evaluation. Across both corpora, four canonical attacker models

anchor the adversarial evaluation track in Table V: Mimicry, Metadata-Strip, Identity-Proxy (GHOST_AUTHOR), and OOD Credential-Phishing. The OOD Credential-Phishing adversary operates during business hours using stolen credentials with low feature novelty, identity-proxy reuse, and removable-media transfers—the regime in which classical novelty-driven insider detectors collapse.

3.2. Attack Vectors and Evasion Strategies

Four evasion strategies are exercised. Mimicry approximates benign-feature distributions to defeat single-layer feature detectors. Metadata-Strip Evasion removes authorship and revision-id signals from OOXML [Didriksen, 2014, Fu et al., 2015], PDF, DWG, EML, git, and image artefacts [Umair et al., 2024, Yassenko, 2025]; this vector specifically targets L1 metadata-only detectors. Identity-Proxy Injection substitutes a clean user's credential surface for the attacker's actions, collapsing naive identity-conditioned rules. OOD Credential-Phishing combines stolen legitimate credentials, business-hours operation, no metadata-stripping, low feature novelty, identity-proxy reuse, and removable-media transfers—the regime in which novelty-, off-hours-, and metadata-driven insider detectors systematically collapse [Ren et al., 2022, Kurniawan et al., 2022]. The four vectors span the limiting cases any deployable insider-threat detector must survive end-to-end.

3.3. Forensic Readiness Requirements

Detection must be coupled to a forensic-evidence chain admissible under ISO/IEC 27043 [Shoderu et al., 2025, 2026]. The architectural target is 100% attribute coverage across the eight mandated attributes: Identification (unique record and alert UUIDs), Collection (documented harvest-pipeline provenance), Acquisition (SHA-256 + BLAKE3 hash verification), Preservation, Analysis (SHAP + GN-Explainer provenance subgraphs), Presentation (analyst-UI workbench export format), Chain of Custody (timestamps and operator-signature tracking), and Integrity Verification (Ed25519 signature over the Merkle root) [Shoderu et al., 2025]. This requirement elevates the threat model from "raise an alert" to "raise an alert admissible as evidence under a forensic readiness audit".

3.4. Detection and Robustness Evaluation Criteria

Detection is reported on three primary metrics: ROC-AUC, F1 @ 0.5% FPR (the precision-critical operating point for low-false-positive deployability), and Matthews Correlation Coefficient (MCC). Calibration is performed via `CalibratedClassifierCV` (isotonic, cv=3); operating thresholds are swept via Youden's J statistic. Target on D3 TSEC: **ROC-AUC 0.9997, F1@0.5%FPR 0.8774, MCC 0.9770**, with the operating threshold pinned at the 0.5% FPR constraint to match the deployment. Target on the D5 OOD_LAB_EXFILTRATOR hold-out: **ROC-AUC 0.9928, F1 0.8936, MCC 0.9539**. Adversarial robustness is bound by **AUC-drop ≤ 5%** against each of the four attacker vectors in §3.2. Cross-layer attribution is bound by a runtime safety invariant: the top-5 SHAP feature importances on the chronological train/test split must

²Project repository: <https://github.com/ali-Hamza817/Turret-OS>

belong to a fixed operational whitelist (access_novelty_score, metadata_stripped, off_hours_multiplier, identity_proxy, copy_to_removable, n_file_accesses, outbound_email, access_hour), guaranteeing zero label leakage at deployment.

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Table 1

Literature Review Summary Table Mapping Key Findings and Structural Limitations Across Prior Work.

First Author	Year	Key Findings	Limitations
Didriksen [Didriksen, 2014]	2014	Forensic analysis of OOXML artefacts establishing that revision-identity fields, custom-XML fingerprints, and document-version metadata can be assimilated into identification pipelines.	Single document format only; no coupled detection model; no held-out adversary evaluation.
Fu et al. [Fu et al., 2015]	2015	Forensic analysis of Microsoft Office 2007–2013 documents demonstrating that XML structure and revision metadata can be exploited; declared useful for covert-channel detection in document artefacts.	Covert-channel extraction orientation; no continuous insider-detection deployment; no held-out attacker evaluation.
Mavroeidis et al. [Mavroeidis et al., 2018]	2018	Data-driven physical-security framework unifying anomaly detection with user-behaviour profiling for enterprise-telemetry insider analysis.	Detection and provenance treated as separate worlds; no ISO/IEC 27043 readiness alignment; no held-out attacker profile.
Li et al. [Li et al., 2021]	2021	Survey of threat detection and investigation with system-level provenance graphs consolidating the provenance-graph paradigm.	Detection and provenance surveyed as separate strands; no cross-format metadata integration; no held-out adversary regime.
Kurniawan et al. [Kurniawan et al., 2022]	2022	KRYSTAL knowledge-graph framework for tactical-attack discovery in audit data using graph-rule fusion.	Alert generation decoupled from forensic-chain emission; no held-out attacker profiles; no cross-format metadata layer.
Ren et al. [Ren et al., 2022]	2022	CSKG4APT cybersecurity knowledge graph for advanced-persistent-threat organisation attribution over indicators-of-compromise, threat reports, and malware artefacts.	Attribution-focused; alerts not coupled to W3C PROV evidence; on-premise coverage absent.
Xiao et al. [Xiao et al., 2022]	2022	MEWRGNN architecture for anomaly-based insider threat detection over user-action graphs.	Single-layer architecture; adversarial robustness not measured; no forensic-readiness integration.
Gao et al. [Gao et al., 2023]	2023	DTGI using Deep Temporal Graph Informax to handle class-imbalance in insider-threat detection.	Delivery decoupled from provenance chain; single-layer GNN without held-out attackers.
Li et al. [Li et al., 2023]	2023	TGCN-DA temporal graph convolutional network with data augmentation for insider-threat detection.	Augmentation over-fit risk on population singletons; no cross-format metadata; no held-out attacker profile.
Wei et al. [Wei et al., 2024]	2024	E-Watcher insider-threat monitoring and detection framework; reporting ROC-AUC 0.9848 on its own benchmark and headline metric accuracy 98.48%.	Single-layer; adversarial robustness not reported; no cross-format metadata fusion; no forensic-readiness audit.
Umair et al. [Umair et al., 2024]	2024	Exif2Vec framework vectorising EXIF metadata to ascertain untrustworthy crowdsourced images.	Restricted to image EXIF format; no extension to OOXML/DWG/EML/git metadata; no detection–provenance coupling.
Xiao et al. [Xiao et al., 2024]	2024	SENTINEL multi-timescale user-behaviour interaction graph learner for insider detection.	Single-layer GNN; missing cross-format metadata layer; no ISO/IEC 27043 readiness alignment.
Aki et al. [Aki et al., 2025]	2025	KGRAG provenance-based cyber-threat-intelligence analysis aligning provenance graphs with retrieval-augmented generation.	CTI-oriented; no continuous internal-detection SLI; no cross-format metadata layer; no forensic-readiness audit.
Yasenenko [Yasenenko, 2025]	2025	Evaluation framework for image-recognition systems for malicious-file detection based on image metadata.	Image-format scope only; no extension to document metadata; no detection–provenance alignment.
Yusof et al. [Yusof et al., 2025]	2025	PROVCON observability-to-provenance framework constructing cyberattack provenance at W3C PROV fidelity.	Detection layer absent; no held-out attackers; no cross-format metadata integration.
Shoderu et al. [Shoderu et al., 2025]	2025	Digital forensic readiness framework under ISO/IEC 27043 for insider-mitigation in SaaS-cloud environments.	SaaS-cloud deployment only; no on-premise/hybrid scope; no detection–provenance coupling.
Shoderu et al. [Shoderu et al., 2026]	2026	Formalised ISO/IEC 27043 forensic-readiness framework for insider mitigation across SaaS deployments.	SaaS-cloud scope only; ML detection not coupled.